Swing Trader · After-Fee Workflow

One question, asked the same way at every stage. **Is there a rule the model can trade that clears the fee on data it has never seen?** Nothing is accepted without passing that test, and nothing is reported that was not computed.

1–2 Data · Frames | 3–4 Screen · Fees | 5–6 Label · Split | 7–8 Fit · Score | 9–10 Verdict · Kill | 11 Book

WORKFLOW CHEAT SHEET

seamusrobertmurphy/swing-trader
Crypto NO-GO · Equities live paper
State of 7 September 2026

1 Where data comes from — EXCHANGE DIRECT

Crypto. The official `data.binance.vision` archive, monthly per-symbol OHLCV zips, checksummed. Not a live API, so a build is reproducible and offline.

Equities. Alpaca daily bars, split and dividend adjusted, from 12,539 active US names screened to 2,673 on a 20M dollar-volume floor. Every request must pass `adjustment=ALL`, or a split halves the series overnight and no error says so.

SURVIVORSHIP COMPLETE The archive listing carries **612 PAIRS** ever listed against roughly 433 alive today, so delisted coins are included. A universe built from today's exchange listing silently deletes every coin that died, and every result on it is an upper bound.

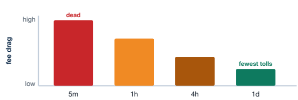
PANELS ON DISK

FRAME	ROWS	NAMES	FROM
5m	5.5M	11	Binance
1h	–	433	Binance
4h	3.9M	567	Binance
1d	520k	540	Binance
eq 1d	5.0M	2547	Alpaca

Stored as Parquet, never CSV. A round trip through CSV turns timestamps into strings and breaks the train and test split without raising anything.

2 Choosing a frame — THE FEE DECIDES

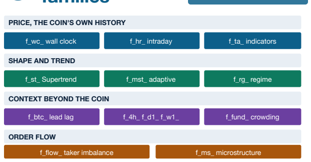
A coarser bar trades fewer times, so it pays the toll less often. That is the whole argument, and it is why the 5-minute frame lost before the model was even consulted.



The indicator's own author says Supertrend bleeds below the 4-hour and holds up above it. The controls layer ruled out 5-minute on the fee wall independently.

COARSER IS NOT FREE Fewer trades means fewer tolls but also less exposure per unit of return. The daily frame did not beat 4-hour; it traded the toll count against per-trade drift in a bear year.

13 Feature families — 104 COLUMNS



Every feature is causal and scale-invariant, so a value uses only bars already closed and means the same on a $90k coin as on a sub-cent one.

TWENTY-FOUR NUMBERS DO THE WORK OF 104 Asked how many bars a trend has left, all 104 engineered features miss by **24.21 BARS** on average. Twenty-four plain lagged returns, with no feature engineering at all, miss by **24.25**. Lower is better and the gap between them is a third of a bar. Both are carrying the same information, so a sequence model, whose one advantage is reading raw history better than a person's summary of it, has nothing left to find.

3 The four-gate screen — ALL FOUR OR OUT

Run per bar, point in time, so a row is kept only if it would have passed on the day. A screen applied with today's knowledge is not a screen.

GATE	THRESHOLD
Liquidity	≥ 30M USDT 24h
Volatility	ATR 2.5–12%
Spread	≤ 0.05%
History	≥ 120 candles

ATR is the average true range, how far a coin typically travels in a day, as a share of its price. The archives carry no order book, so the spread is estimated from each day's high and low by the Corwin-Schultz method. The live screen uses the real spread.

RANK WIDE, TRADE NARROW The universe is scanned broadly, then a whitelist of coins that actually carried the out-of-sample edge is enforced at entry.

4 What a trade must clear — THE FLOOR

Net move, never gross. A trade is refused unless what is left after the round trip clears the floor.

```
# the whole fee argument in one line
net = move − round_trip − slippage
allow if net ≥ edge_floor
```

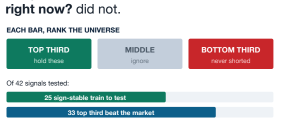
0.15	0.20	1.5
% achievable round trip	% standard round trip	% minimum net edge

0.075% per side with the exchange token discount and post-only maker entries, so no spread is paid. The live wrapper reads the account's **actual** rates each run and warns when the measured round trip exceeds the modelled one.

THIS IS THE WALL A basis point is a hundredth of a per cent. Crypto's real edge came out 5 to 12 of them *smaller* than the 15 to 20 it costs to open and close a trade. No gate, label, frame or feature family closed that gap.

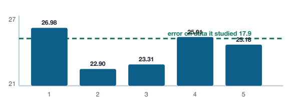
14 Cross-sectional — RANK, DO NOT PREDICT

Two different questions. **Will this coin rise?** failed everywhere. **Which of these coins is strongest right now?** did not.



TWO STATEMENTS, KEPT APART The ranking signal is real and was never disproved. The BTC-gated *way of trading it* was killed at 27% of folds against the 60% bar. Those are different claims.

23 Reading the fold chart — SPREAD BEATS MEAN

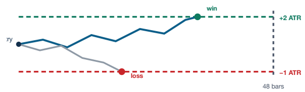


Bars near the dashed line generalise. Fold 1 is worse than guessing the average.

NEVER READ THE MEAN ALONE These five average to the same number a steady 25 would.

5 The label — TRIPLE BARRIER

What counts as a win is a decision, not a fact. Three barriers settle it: a profit target, a stop, and a clock. Whichever is touched first is the answer.

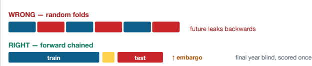


Barriers scale with ATR, so a calm coin and a violent one are judged on the same terms rather than the same percentage.

A LABEL CAN LOSE BEFORE THE MODEL SPEAKS On the 4-hour panel the profit target is reached first on **269 OF EVERY THOUSAND BARS**. To break even against a stop set half as far away, it would need **333**. Negative expectancy is built in, so no model on it can win. Check the base rate against breakeven *first*.

6 Splitting time — NEVER AT RANDOM

A random split puts next Tuesday in training and scores the model on last Monday. Market bars are autocorrelated, so that leaks the future backwards and flatters everything.



Embargo is a gap at the cut, as wide as the label horizon, so no label straddles it. Without it the last training rows carry outcomes that live in the test window.

15 Model scorecard — FULL VS CROSS VALIDATED

MODEL	FULL MAE	FULL RMSE	CROSS VAL. MAE	CROSS VAL. RMSE	RATIO
HistGBM	.2181	.2619	.3861	.4818	1.840
LightGBM	.2387	.2806	.4036	.4826	1.720
RF	.4445	.4526	.4761	.4840	1.069
LogReg	.4718	.4855	.4463	.4909	1.011

4-hour frame, 7 Sep 2026. The error here is measured on the probability the model states, not on a yes or no call, so a model that says 80 per cent and is wrong is punished more than one that says 55 per cent and is wrong.

THE WINNER FAILS THE RULE Selection takes the best cross-validated RMSE and never consults the ratio, so it crowned HistGBM at **1.840** while the bar rejects anything over 1.1. The two stable models sat 2 to 5% behind on error.

7 The four numbers — READ TOGETHER

RMSE average miss, big misses squared first so one disaster moves it

MAE average miss, every miss counted once

MAPE miss as a share of truth; needs a target that is never zero

ratio CV RMSE ÷ train RMSE

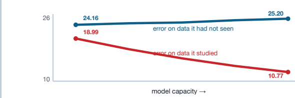
When RMSE far exceeds MAE the model is **occasionally very wrong** rather than steadily slightly wrong. That distinction changes what you fix.

THE RATIO WAS UPSIDE DOWN UNTIL 7 SEP 2026 It divided train by CV, so an overfit model scored 0.91 where the rule expects 1.10, and anyone applying "reject above 1.1" passed exactly what it exists to catch.

≤ 1.1	> 1.1
stable, keep	memorised, reject

8 What tuning showed — LESS, NOT MORE

Nine settings, three walk-forward folds, 4-hour frame, predicting bars until the trend flips.



Training error nearly halves while held-out error gets worse. Every one of the nine settings is rejected on the 1.1 bar, and the winner was the weakest model in the grid.

PERIOD BEATS EVERYTHING, 40 TO 1 In the lag comparison, three ways of feeding the model differed by 0.094 bars. The same model asked about different stretches of time differed by 3.83. Which months you test on matters forty times more than how you build the model.

16 The calibration hole — NOTHING BUILT YET

Every model trains with **balanced class weights**, which deliberately pushes predicted probabilities toward 0.5 and away from the true base rate. The headline metric is Brier, which measures exactly that fit.

Drawn here to show the shape, not measured. Nobody has yet checked whether a stated 30 per cent chance happens 30 per cent of the time, and nothing corrects the numbers afterwards so that it does. That is the piece of work still to build.

WORSE THAN A CONSTANT Always predicting the base rate scores **0.4155** on daily; the best model scores **0.4662**. Twelve per cent worse. It does not mean no skill, it means the probabilities are wrong even where the ranking is not.

9 Passing the bar — FOLDS, NOT TOTALS

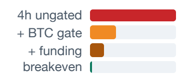
A total can be carried by one lucky stretch. History is cut into half-year folds and the edge must hold in **60 per cent** of them, on two questions at once.

TEST	ASKS
Absolute	does the picked cohort make money after fees
Selection	does it beat the market in the same fold

ABSOLUTE ALONE IS A TRAP Markets drift up, and a panel holding only the coins that survived drifts harder still. A rising tide and a flattering sample can produce a profit between them with no skill involved, so the real bar is beating the market inside the same fold.

SURVIVES	PARTIAL	KILLED
both bars, kept	closer, not clear	falsified

10 Where it stands — HONEST SCOREBOARD



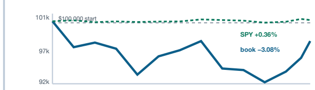
Per-trade return after fees, out of sample. A year of work moved the line from −0.38 to −0.05, and none of it crossed zero.

THEY LOSE BEFORE FEES Edge diagnostics, 7 Sep: daily −1.150% and 4-hour −0.182% per trade *pre-cost*. That reframes a year of calling this a fee-wall problem.

THE ONE LIVE EDGE

12-1 equity momentum. Buy what rose over the last year, skipping the most recent month. Over 116 months it beat the average stock by **1.006 percentage points a month**. That is 2.41 times the size of its own wobble, which is the usual bar for calling a result real, and it held in 85 and 80 per cent of the half-year folds. Paper book live since 18 Aug.

17 The live book — PAPER, REAL PRICES



Fourteen sessions. The plan expects swings of about 8.2% over that span, so a 3.1% dip is 0.4 of an ordinary move rather than a signal.

50	3/6	7.3
names @ 1.8%	clean cycles	bp per fill mid-session

THIS IS FAKE MONEY A paper account at real prices. `LIVE_TRADING` is off and every order path aborts unless it is the exact string `true`.

20 Before deploying — ALL EIGHT

A check is finished when all of these hold. Not before.

- **Base rate beats breakeven.** Check the label's own expectancy before any model is fitted.
- **Split is forward-chained** with an embargo as wide as the label horizon.
- **Blind window scored once.** A threshold chosen by looking at it is no longer blind.
- **Both fold bars cleared,** absolute and selection, at 60 per cent or better.
- **Overfit ratio at or under 1.1,** and selection consulted it.

11 Traps that cost days — ALL SEEN HERE

- **Rendering is not checking.** A page can build, exit zero, and report 0 of 6 cycles instead of 3. Compare against the previous render.
- **Gitignore does not untrack.** A rule only stops files git does not already track. `git ls-files -i -c --exclude-standard` lists what needs removing.
- **Paths built from pieces.** A search for `"outputs/"` misses `join(x,"outputs",y)`. That wrote 51MB to a stray folder, twice, silently.
- **Balanced weights break Brier.** Class balancing pushes probabilities off the base rate while Brier measures exactly that fit. Every model scored worse than a constant.
- **A timeout hides everything after it.** One slow cell aborted a run and concealed 20 cells behind it.

12 Running it — SCRIPTS, NOT NOTEBOOKS

```
# weekly, intraday only
alpaca_data.py download
alpaca_trade.py rebalance --execute
alpaca_trade.py check
alpaca_execution_report.py
```

NEVER OUTSIDE THE SESSION Orders queued into the opening auction cost **42.6BP** per fill against **7.3BP** mid-session. Same book, eleven times the cost, from the clock alone.

HARD RULES IN CODE

5%	3%	10%
position cap	daily circuit	cash floor

Never short, never margin, never options, never average down. Enforced in code, not assumed from the venue. `LIVE_TRADING` must be the exact string `true` or every order path aborts.

18 Dropped features — CLOSED ARTEFACTS

DROPPED	WHY
1h direction	AUC 0.50, coin-flip floor
Entry sharpening 4h	29% of folds, regime artefact
MST + breadth gate	27% of folds vs 60% bar
Funding gate	−0.067% at achievable cost
Daily frame	label degenerate, base rate 0.068
Microstructure	no selection edge
5m scalp	fee wall
Sequence models	0.25% gap, priced not worth it

A KILL IS A RESULT Each one is written up with its numbers and kept. The harness was validated by killing 6-1 momentum and low-vol exactly as the literature predicts, which is why its verdicts are trusted.